Turtle Games: Business Analysis

Business analyst: Mia Sasunic

Analysis approach:

1. Background and descriptive statistics

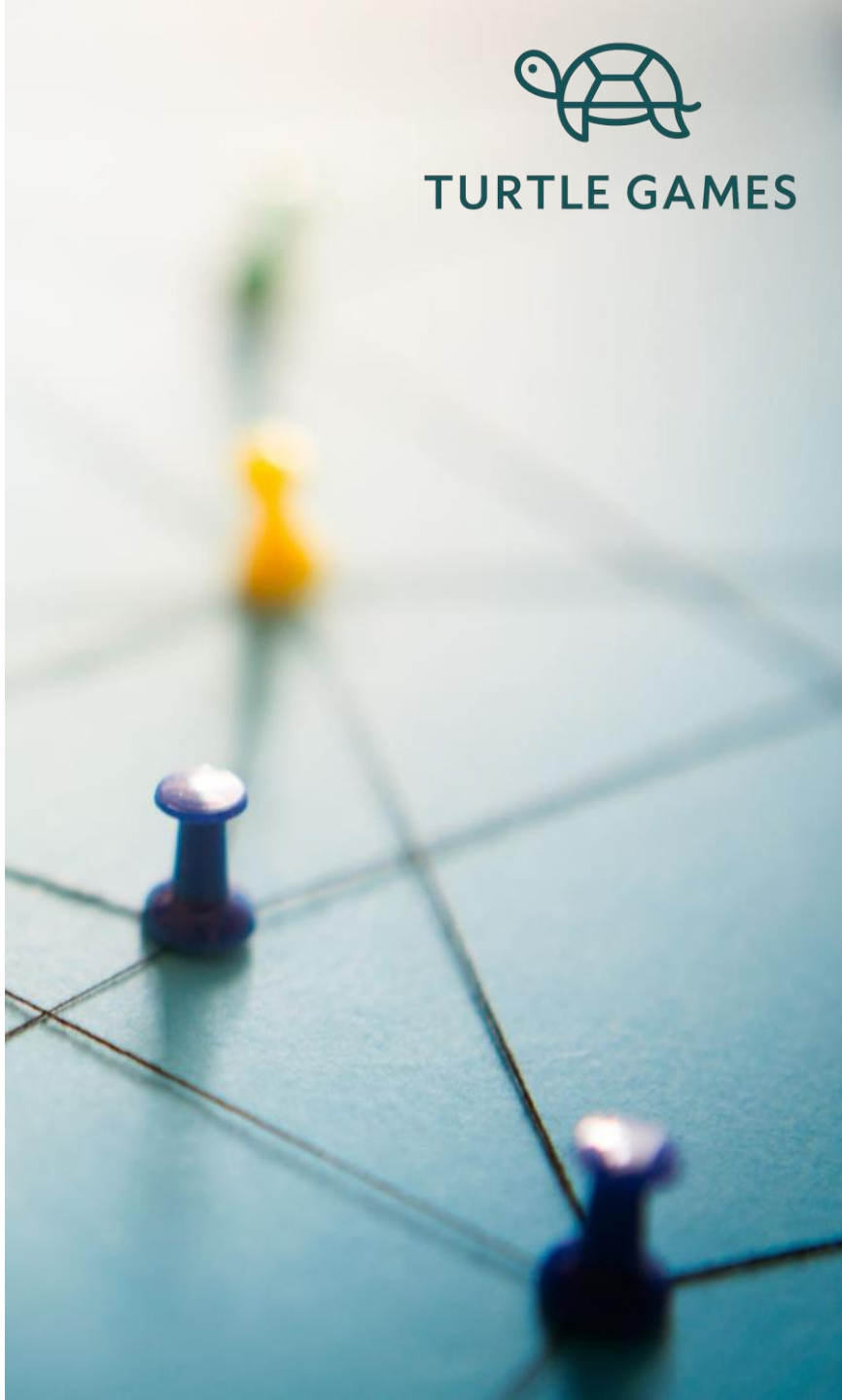
2. How customers accumulate **loyalty points**?

3. Definition of **clusters** in customer behaviour

4. Sentiment tracking in marketing campaigns:

   Benefits of **social data** (e.g. customer reviews)

5. **Recommendations** and conclusion



Background and context

- **Business goal:**

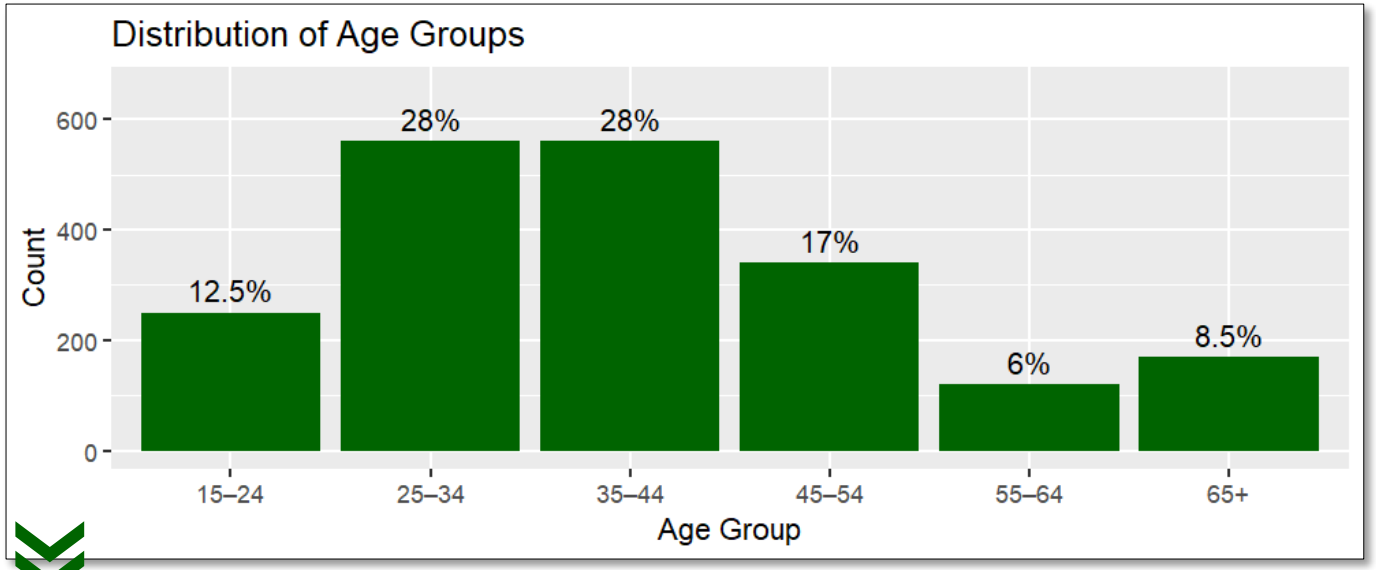
  Improve overall sales performance by analysing and considering customer trends, specifically loyalty score drivers

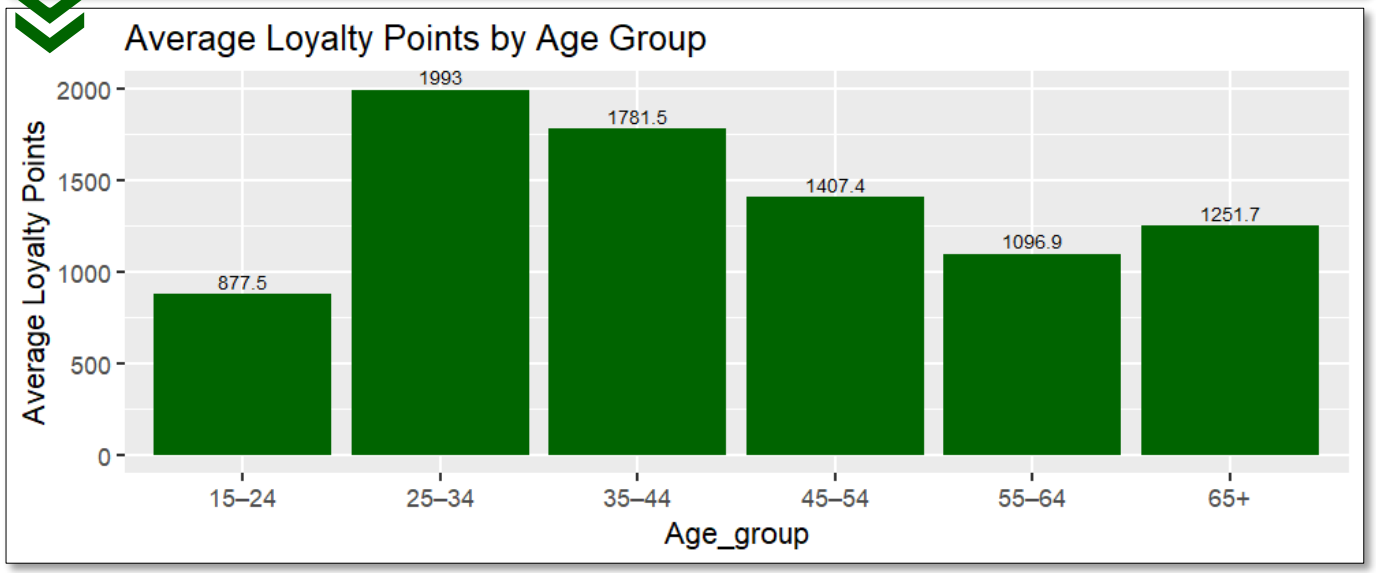
- **Data source:**

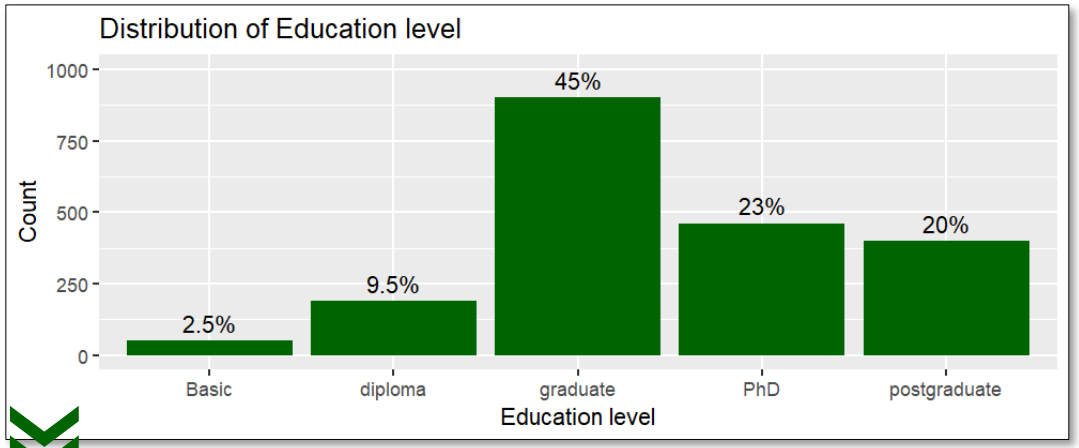
  2000 customer reviews including categories:

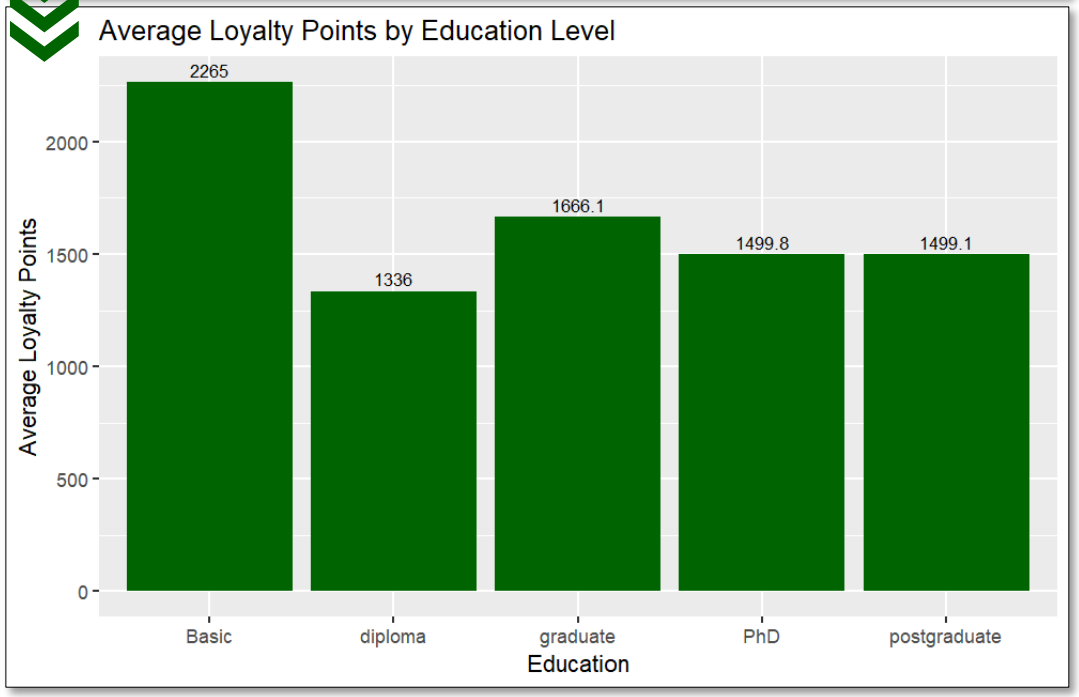
  - Gender
  - Age
  - Remuneration
  - Spending score
  - **Loyalty points**
  - Education
  - Product
  - Review
  - Summary

Customer distribution vs. average loyalty score: Most-represented age groups (25–44) also show the highest loyalty

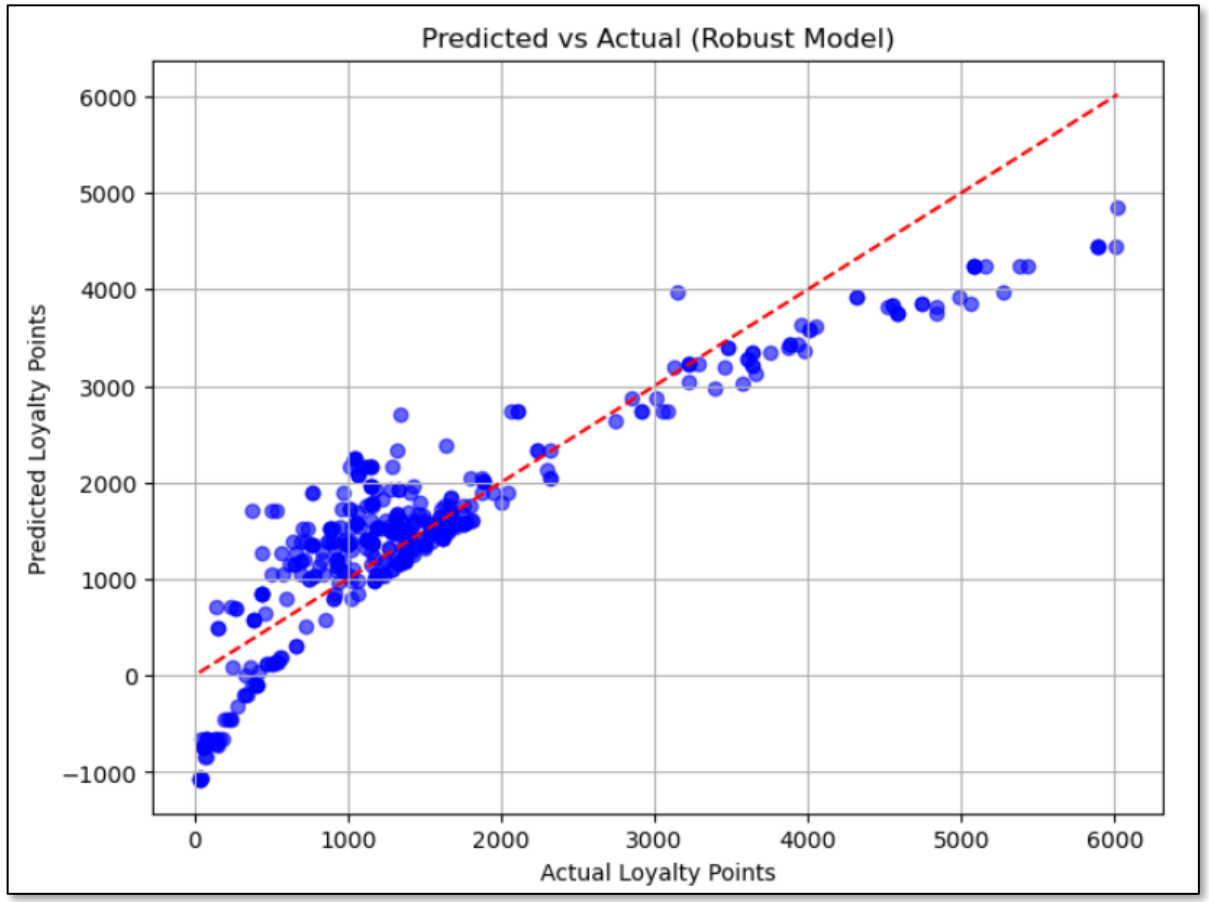








Remuneration and spending score significantly influence loyalty points



Multiple regression analysis

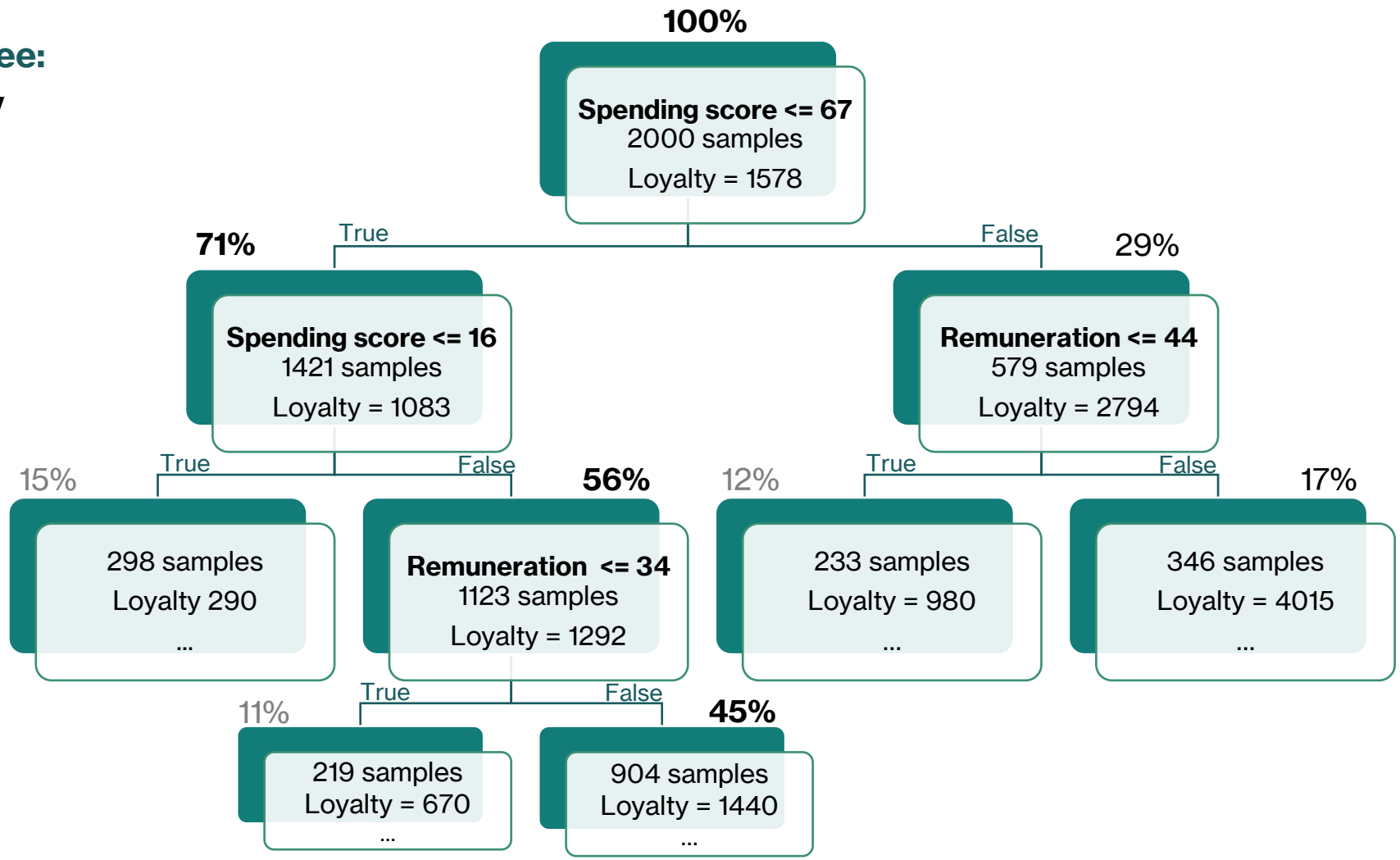
→ model with **83% probability**:

- when **remuneration** increases by 1, loyalty points **increase by ca 34**
- when **spending score** increases by 1, loyalty points **increase by ca 33**
- Other variables not significantly relevant

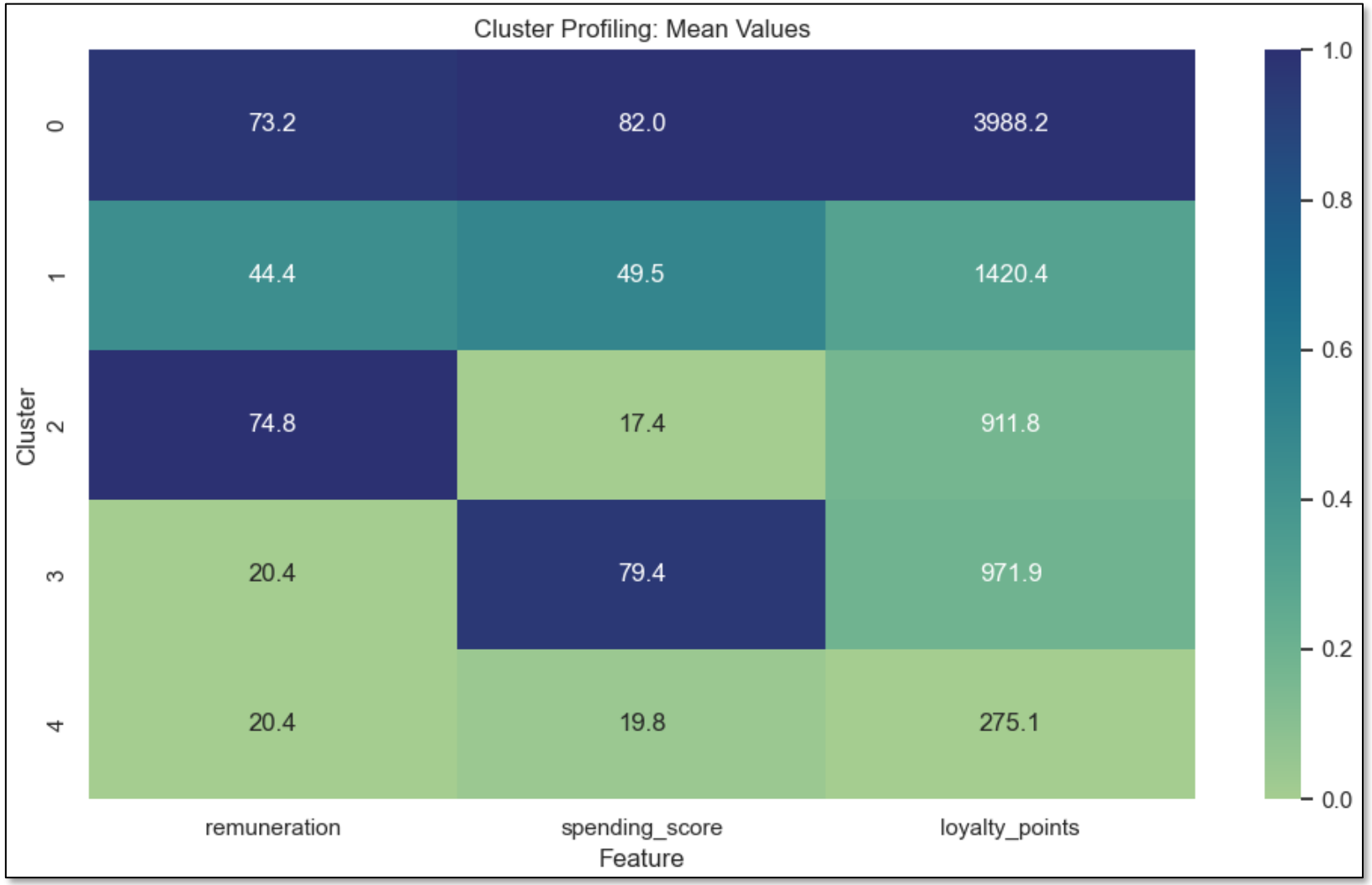
Most relevant customer group has spending score between 16 and 67 and remuneration > £34k

Excerpt from Decision Tree:

→ model with **95% probability**



5 customer groups were identified with K-means clustering method



Group 0 – 356 samples
- High income
- High spending
- ➢ **High loyalty**

➢ **Premium buyers**

Group 1 – 774 samples
- Average income
- Average spending
- ➢ **Average loyalty**

➢ **Steady buyers**

Group 2 – 330 samples
- High income
- Low spending
- ➢ **Low loyalty**

➢ **Detached buyers**

LOW / AVERAGE SPENDING

Group 3 – 269 samples
- Low income
- High spending
- ➢ **Low loyalty**

➢ **Impulsive buyers**

Group 4 – 271 samples
- Lower income
- Low spending
- ➢ **Low loyalty**

➢ **Budget buyers**

Customer review text analysis shows strong positive sentiment, though scoring is inconsistently assigned

Excerpt from Top 15 negative comments:

Comment ID	Review Text	Score
REVIEW #280	I bought this thinking it would be really fun but I was disappointed its really messy and it isnt nearly as easy as it seems also the glue is useless for a 9 year old the instructions are very difficult	-0.9520
REVIEW #1053	for those just getting started in the dnd world or those who havent played dnd in years if not decades this is a good way to get learn, reacclimate yourself to the games played an adventure with a friend on the first day while the amount of pieces tokens tiles was a little overwhelming at first got accustomed to the game pretty quickly and had quite a bit of fun my only con about it is that ashaldaronthe figure seemed well it had great detail on it but the **assembling in factory seemed bad or poor there is a slight gap between the two sections** and what appears to be a slight overflow of glue that was used i guess i kind of expected a one piece mold or if not a bit more care taken when assembling i do plan on painting these figures at some point and im not sure whether this is normal or not as well as whether it will be possible to really paint this particular figure without it looking off due to the gapglue problem perhaps i got a bad piece ill probably end up talking to customer service about this issue and see about getting **a replacement** that was assembled properly	**-0.9539**
REVIEW #889	the game is amazing the new version is not cardboard is biodegradable but a greener version could have been produced with better game pieces i guess avalon hill has beaten us all at its own board game they no longer need to cater to the consumer because people will buy even **though this version is terrible in comparison to the 1999 edition** disappointing but not surprising	-0.8984
SUMMARY #1453	Adds a needed amount of tension in a tense fantasy world	-0.5719
SUMMARY # 172	Horrible Nothing more to say Would give zero stars	**-0.5423**
SUMMARY #1508	Very cute but poor quality stickers	-0.5266

Despite its limitations, NLP enables scalable insight extraction from lengthy customer comments

Excerpt from Top 15 positive comments (for illustrative purposes):

1105	disclaimer bought this from a local store paid list value but supporting local game stores helps keep them in business and its a rough market to keep a gaming store running also i do not own nor have i playednbspa datahookproductlinklinked classalinknormal hrefdungeonsdragonscastleravenloftboardgamedp0786955570refcm_cr_arp_d_rvw_txtieutf8dungeons dragons castle ravenloft board gamea the first dd adventures board game regardless dungeons dragons wrath of ashardalon board game woa is a fantastic value at 65 list you get 42 plastic unpainted minis 1 scale you get a nice set of cardboard interlocking dungeon tiles and you get a fun very random board game playable either by yourself or with up to 4 friends what it is woa is in gamer terms a roguelike in tabletop form in laymans terms woa is a dungeon crawler you start at the entrance and have a specific goal rescue the villagers survive the dungeon destroy the altar slay the dragon the adventures are built randomly based on shuffling the tiles and they are played individually as you explore you can go for a massive linear dungeon based on a little luck and careful planning or you can attempt to condense yourself in a larger swarm interconnected tunnels some of the adventures feature chambers special larger rooms that are all placed at once and filled with enemies and a specific goal that the team must complete to win the adventure its a great way to introduce someone into tabletop roleplaying gaming its the perfect medium between a board game and an actual pen and paper rpg its a fun quick game to run a quick adventure for some friends taking anywhere between 20 minutes and an hour per adventure what it isnt a deeper roleplaying game with a full interconnected series of quests over a grand campaign a completely accurate representation of 4th edition dungeons dragons rules overly complicated as most roleplaying games will seem to be for most people whats in the box 42 plastic heroes and monsters 13 sheets of interlocking cardstock dungeon tiles 200 encounter and treasure cards rulebook scenario book 20sided die miniatures the game includes 42 minis including 7 villan 5 hero and 30 monsters the minis are of nice quality theyre clearly just mass produced but they put some effort into modeling them just not so much into trimming the leftover plastic in places some of the swords will be bent from being in the box and theyre unpainted however i dont mind the unpainted state and i plan on using them during dd encounters im dming at a local store the ashardalon figurine is very detailed my only and the only real gripe is that none of them are painted 45 cardstock dungeon tiles theyre one of the nicest parts of the set theyre very high quality the art is nice and the interlocking design cant be beat purchasers of dd dungeon tiles will know how hard they are to keep together without tape these lock together like puzzle pieces wouldnt be hard to use them in a real dd game except for the fact that theyre somewhat limited and theyre not large open spaces there are a lot of tokens and hp markers and other things all made out of the same cardstock including cards for the villans your big bad guys and the heroes 55 200 encounter monster treasure cards these are drawn during the different phases of the game there are 3 and they certainly add to the randomness of the game everything scales nicely so that you can attempt to run the party adventures solo and itll still work nicely 55 rulebook the rulebook is the only real downfall ive seen of the box set it does an alright job at conveying the basics but other questions i have are just nonexistant can i spend a healing surge without being dead just to heal the book itself has card errata on the back page which is crazy other than that its not a bad rulebook for being only 15 pages 355 scenario book adventure book this has 13 adventures to run using the set each of them varying based on how things shuffled what magic items get drawn and if theres a chamber what chamber card gets drawn the adventures essentially play themselves as you follow the turn orders theres very little fluff for each of the adventures but thats what make it dd you get to set your own motivation for the adventure your own background its enough to get you going and you let your imagination take it from there 455 20 sided die aka the d20 the iconic dd symbol you roll it add a modifier and compare it to a set number thats pretty much it its just as nice as the dd red box dice or any chessex dice ive bought but its just a d20 nothing special nice if you dont have any throw it in the bag with all your others 55 overall i would have liked to see more adventures thats not to say more arent coming the official ones available at time of review require castle ravenloft as well which i dont have yet theres only one solo adventure but i played through it and adventure 5 solo not sure if a single player could take on ashardalon himself but half the fun is seeing how well you can do i actually wouldnt mind running a full dd campaign with similar designs its a perfect format for a megadungeon delve its fast and its exciting due to the randomness im glad the cards arent collectable theres 200 thats all there is all in all if youre into rpgs or if youve ever wondered if a roguelike could work in a tabletop format it can and its called wrath of ashardalon edit i forgot to mention the initial setup punching out the cards takes a considerable amount of time like close to 20 minutes setup per adventure doesnt take nearly as long maybe 23 minutes	0.9996
869	whenever i see this game on my shelf i get a disturbing visual of quarks big head from star trek deep space nine i then picture him playing tongo with a bunch of other ferengia game that deals with buying selling acquiring and other things youd normally see on the wall street floor dont ask me whyi havent had my coffee yet this morning acquire 36 players ages 12 average play time 60 to 90 minutes note while the box says that the game is for three to six players i dont see why a two player variant couldnt existif one doesnt already im fairly confident that two players with an imagination who are not concerned about following the rulebook to the letter could make do i also came across the older avalon hill versions rulebook and there is a section for special rules for two playersso feel free to experiment acquire plays a bit differently than the other games ive played the closest comparison i could possibly make to what ive already reviewed is airlines europe in both games youre buying stock and whoever has more of certain stocks colors often reaps the biggest reward when payouts occur more on that in a minute version components the version of acquire i purchased is the latest print of the game by wizards of the coast i need to get this pet peeve of mine out of the way before i begin the player tile trays are made of fairly cheap cardboard that you have to punch out and bend to make the pieces assemble in other words i had a fun time trying to assemble the player tile trays in most games ive purchased tile trays came assembled and were of much better quality some people may not mind the extra work and the game is fairly cheaper than your average eurostyle board game but cmonwould it have hurt profits that much to just make the components a little better also the player reference tables were part of the manual itself that i had to cut tear out for ocd people like me the manual just doesnt look right now that pages look torn out of it i dont see why they couldnt have printed reference sheets separate of the manual previous versions of the game were much nicer looking and of better quality so much in fact that other reviewers recommend trying to find those older printed versions as opposed to this one i dont mind average to poor production values if the price is right but this is just pushing it game mechanics the game board is made of up of squares that form a grid going from 1a in the upper left hand corner all the way to 12i in the lower right hand corner to the left of the grid are the available corporations that players can form and to the right is the same reference chart youll find in the cutouts in the back of the manual in acquire turns are fairly simple players place a tile players buy up to three stocks and finally players pick up a new tile from the supply tiles correspond to the squares grid on the boardfor exampleyoud place the 1a tile if you had it onto the 1a square on the board grid when a player places a tile next to an existing tile sitting by itself that player gets to form a corporation it doesnt become theirs mind you but they do get a free stock card for establishing it that player can choose any of the available corporationsthough some grow differently than others some corporations are cheap to buy stocks for early on but dont pay out as much when they grow in size while others are expensive to buy stocks for early on but payout is big in the endgame players continue placing tiles forming growing corporations and buying stock cards until a tile is placed that causes two corporations to bump heads this is the real meat and potatoes of the game a few things happen when this occurs 1 the larger corporation acquires or eats up the smaller one if the purple corporation for example had six tiles and green had three tiles purple would acquire green the exception to the rule is if both corporations are eleven tiles are higher in size corporations with eleven tiles or more are considered safe 2 the player who owns the most stock cards in the company that was acquired gains the one time majority shareholder bonus the amount varies depending on how big the corporation actually is the player who owns the second most stock cards of the company that was acquired receives the one time minority shareholder bonus sorry but there isnt a prize here for third place unless there is a tie for second place 3 players who own stock in the company that was acquired have a choice to make they can sell their stock for cash now the amount depending on how big the corporation is they can trade that stock on a 2 to 1 basis for stock of the company that acquired it they can keep their stock hoping the corporation will form again down the line the acquired company marker gets put back to its place on the side of the board and can be formed again later on at the end of the game shareholder bonuses are given out to the companies that are on the board and their stocks are traded in for cash the winner is the player with the most money after that is all said and done the review all three kids joined me to form a four player game explaining the rules wasnt as bad as i thought it would be i was concerned that the whole buying stocks and acquiring would overwhelm the eleven year old but ironically he ended up winning the game he randomly bought stock so that hed try to have more than everyone else which ended up working out to his advantage unbeknownst to him i focused on buying stocks for two corporations in the middle of the board and would have made a lot of money had one acquired the otherbut they ended up becoming safe and i didnt have much money left to buy stocks in other corporations that could still be acquired i received a nice bonus at the end of the game for all the stock i had but my son used the money he earned from his acquisitions to buy even more stock than what i had bought up cheaply in the beginning of the gameso he ended up getting the majority shareholder bonus anyway he earned so much from a particular acquisition that he just started buying out all of the stock cards of particular colors until there were none left it seems to pay to have money throughout the game and not to focus on one or two corporations unless you are sure they will pay out i was tempted to offer him a later bed time if hed slip me ten grand but i thought better of it the game took us about two hours to playkeep in mind there was a learning curve and three kids of varying ages involved overall i really enjoyed the game acquire is simple enough to learn but deep enough that players can form strategies in order to outdo the other the random tile placement throughout the game by players keeps the replayability high i quickly forgot about the quality of the player tile trays when i started playing out scenarios in my head on how certain corporations might play out on the board the kids were actively working out aloud which corporations they planned to grow and were constantly comparing how many stock cards they had of certain colors they were often on the edge of their seats when a merger occuredanxious to see how it would play out the older ones took it a step further and bought out stock in the growing corporations recognizing how much money they could make when it grows even bigger the younger one just bought what he could afford and random chance worked in his favor they all stated that they enjoyed playing the game i personally recommend acquire if you enjoy games with a financial theme to them in this case stocks and money i recommend that players have a calculator handy especially for the endgame the poor production values when compared to versions previous to it might turn some people off but id recommend putting that aside in your mind to enjoy the game for what it really is	0.9995
1636	if you only employ one creativityenhancing resource for the rest of your life make that resource the ball of whacks breakthroughs in effective creativityinspiring methods seldom occur most new methods are simply restatements or reshapings of ideas that flourished among the ancient greeks such as heraclitus a favorite source for mr von oech or before those who are familiar with the earlier work usually dont gain much from the newer approaches the ball of whacks is a happy exception heraclitus would say that you can never interact with the ball of whacks twice in the same way and he would be right the most valuable creativityenhancing methods ive experienced usually require having a team work together with one of the tasks involving making little models or prototypes of creative concepts i often find myself wishing i could have those experiences more often but seldom having the right people and time to pursue that desire then i discovered roger von oechs ball of whacks wow my frustrations are now behind me this is day 8 of working with the ball of whacks its been quite an education i can hardly wait to find out what ill learn on day 9 when you first see the ball of whacks you wont quite know what to make of it open the box take out the ball the magnetic stand and the workbook then let yourself go discover what the ball is to the geometrically inclined youll notice that its a rhombic triacontahedron try saying that 5 times fast a balllike shape with edges formed by 30 rhombi a rhombus is a foursided equilateral with two pairs of parallel sides that resemble scaleddown versions of an egyptian pyramid done in red the ball feels somewhat alive however it gives when you squeeze it and resumes it shape when you stop squeezing it as soon as you take out a piece youll discover that the ball is composed of 30 design blocks that use magnetism to adhere to one another as you take the pieces apart the ball disappears and many regular and irregular shapes occur usually thats the end for me i can never recover the original shape my rubiks cube is still a mess but the ball is easy to reform if i can do it you can too i like abstract shapes and i was pleased to find that the magnets act both to attract and repel the pieces unlike lego pieces connections can be made in all kinds of odd ways i also noticed that the pieces often reshape themselves almost organically as they choose a different connection than the one that i initiated the casual results are often better than my planned ones based on a suggestion in the workbook i began mixing in other objects that could be held by the design blocks the results were infinitely more interesting and rewarding than anything id ever created before i felt a great sense of peace from the experience many people had told me i would like sculpting but i lacked the confidence to try with the ball of whacks i felt like a super sculptor but without any lessons after several days of working on abstract shapes i decided to try geometric shapes these were paradoxically harder to do in many cases the magnetism sometimes fought me i had to design clever solutions to create the regularity i wanted to see i also left the ball of whacks out to see what others would say everyone noticed that i kept changing it without thinking rewarding new designs would emerge in a few seconds i gained confidence in my creativity involving forms at the same time i was working on a number of nonphysical problems soon i began seeing metaphors in the blocks and my designs for those problems new and better solutions began to flood my mind feeling like i should read more of the workbook i now found myself intrigued by the directions for how to use the ball of whacks to expand my creative repertoire i began to see huge gaps between the methods i prefer to use gaps that i could easily explore and understand with my new appreciation of my creativity habits best of all the lessons are deeply ingrained the human nervous system is mostly made up of connections between the hands and eyes to and from the brain by both holding and seeing what i was working on the exercise connects more parts of my brain and experience i find totally different memories being stimulated especially those from the sandbox and while playing with modular toys today i went back and totally redid a book proposal im working on i was astonished to see that my mind was connecting those little blocks from the ball of whacks into new ways of assembling the pieces of the proposal i would never have expected that what does the name mean i dont think theres just one meaning it obviously sounds like ball of wax which is something that warms and becomes malleable in our hands im reminded of the red wax that you peel off of edam cheese to form shapes at another level a whack in von oechs vocabulary is a way of getting you to let go of artificial assumptions in that sense this is a tool to give you lots of whacks and it does i havent done all of the exercises yet but i plan to each of the ones ive done has been quite an eyeopener heres a tip turn off the tivo while youre watching television and use the commercial breaks to play with the ball the baseball playoffs are excellent for this purpose because you get all kinds of short delays be smart give a ball of whacks to each of your children rather than purchasing another toy youll inspire genius that will constantly fascinate you	0.9992
	as a dad of two boys im always on the lookout for activities for us to do together something we can all enjoy and equally get into we built a da vinci catapult a siege tower did some exploring with microscopes geocaching etc with varying degrees of interest in one of the two boys dont get me wrong we all had fun but it was more for one or the other depending on the activity my older son since the day he could has loved to talk he loves to tell stories about things hes done things hed like to do or things hes watched if he can make a story out of something he will my younger son has since i can remember loved to play board games there are ones he likes more than others but i dont think weve played a game he didnt like i recently became aware of a cardboard game by a company out of the uk named cubicle7 the game is called hobbit tales from the green dragon inn and it looked very interesting cubicle7 also makes a rpg that takes place in the same world as this game called the one ring the cards in this game can be used with that rpg if desired so its all very integrated the basic premise of the game is that the players are hobbits sitting around a table at the green dragon inn telling stories of their adventures the game takes place in rounds one for each player and they tell the story with the aid of adventure cards 75 of those each adventure card has a story aid something to inspire the story teller a title and a tolkien quote along with some excellent artwork to really inspire you however the other players have similar cards called hazard cards 40 of those that they use to try to throw the story off things like a goblin patrol or a group of brigands this is the real magic of the game when the players interact with each other in the telling of their or each others story the goal of course is to score the most points of course but the fun isnt	

Recommendation: Increase customer spending to boost loyalty (1/2)

PRIORITY A customer segments:

Detached Buyers (17%)

- **Traits: High income | Very low spending | Low loyalty | Mostly aged 35–44 | Mostly highly educated**

➢ **Re-engage through premium experiences:**

- Offer exclusive previews of upcoming releases
- Provide early access to limited-edition products
- Send premium sample items (miniatures, demo packs)
- Run storytelling campaigns highlighting product quality and brand values

Steady Buyers (39%)

- **Traits: Average income, spending & loyalty | Mostly graduates | 60% females**

➢ **Grow basket size among already engaged customers:**

- Use personalized recommendations based on past purchases
- Promote limited-time bundles (e.g., game + expansion)
- Suggest cross-sell items (accessories, add-ons)
- Share educational content (how-to-play guides, strategy tips)

Recommendation: Increase customer spending to boost loyalty (2/2)

PRIORITY B customer segments:

Premium Buyers (18%)

- **Traits: High income & spending | Very high loyalty | Mostly aged 25–44 | Mostly highly educated**

➢ **Reinforce premium loyalty:**

- Provide VIP early access to new releases
- Offer exclusive loyalty-tier perks (collectibles, digital bonuses)

Budget Buyers (14%)

- **Traits: Low income & spending | Very low loyalty | > 50% aged btw. 35-54 | Mostly highly educated | >60% females**

➢ **Maintain at low cost:**

- Use automated seasonal discounts
- Offer occasional low-cost promotions (clearance sales, budget bundles)

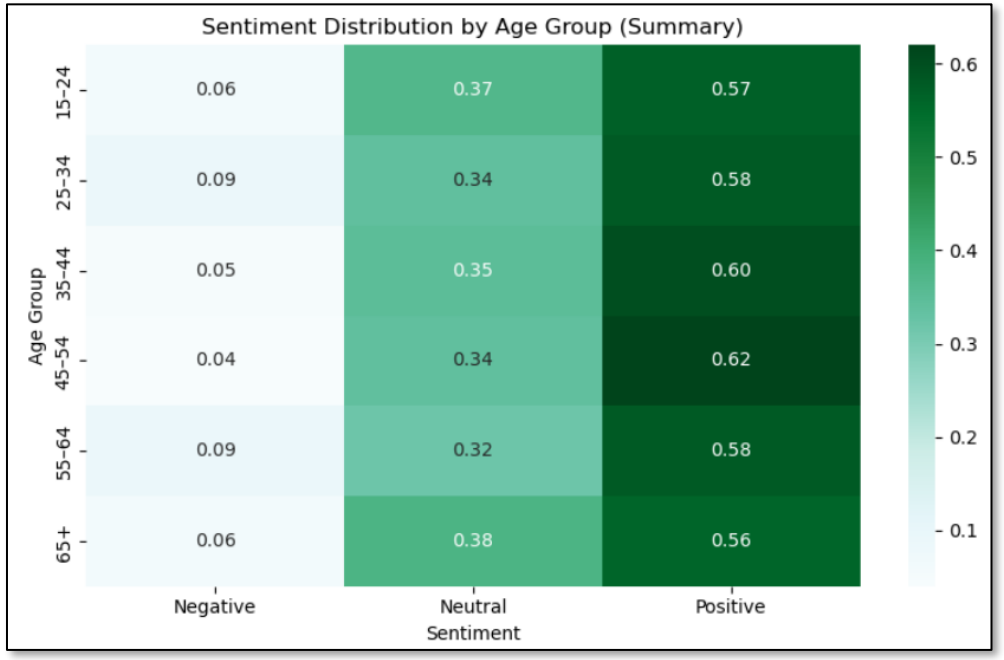
Impulsive Buyers (13%)

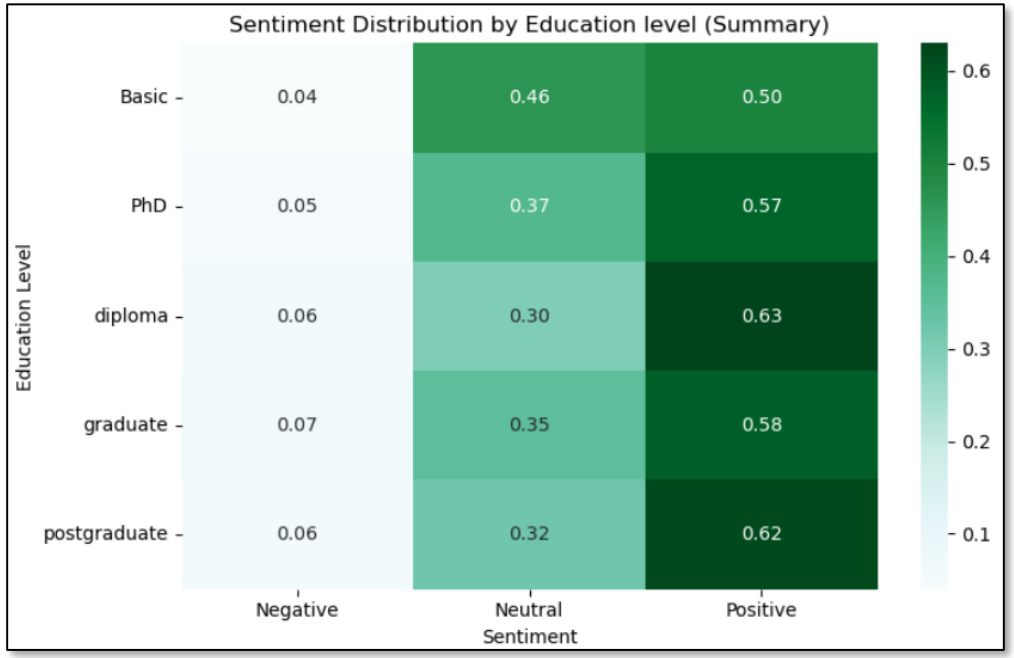
- **Traits: Low income & loyalty | High spending | Mostly 15-34 | Mostly highly educated**

➢ **Encourage Consistent Purchasing:**

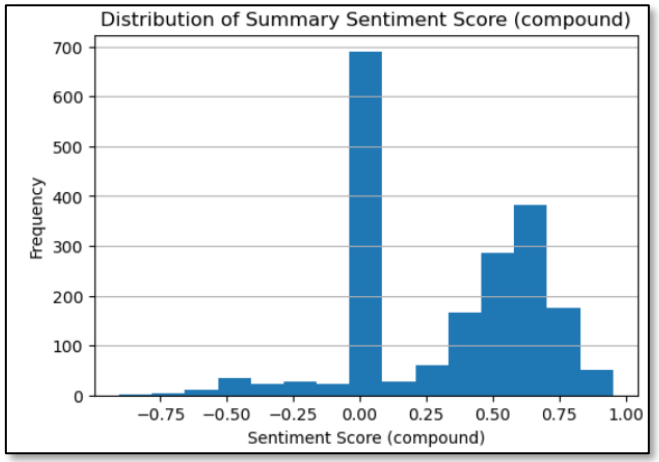
- Introduce delayed rewards (“Buy 2 this month, get 1 accessory next month”)
- Reward consistent monthly purchases (streak-based loyalty boosts)
- Use “complete your set” nudges for collectible lines

Additional insight: Overall positive product sentiment with isolated quality issues





- **Definite positive trend**, especially among age 45-54 and higher levels of education
- Negative comments mostly related to product numbers: 5429 (4x), 9119 (3x), 466 (3x), 10196 (3x), 1241 (3x) commenting product quality → **check internally**
- Limitation: most comments classified as neutral, scoring is inconsistently assigned -> **tool improvement necessary**



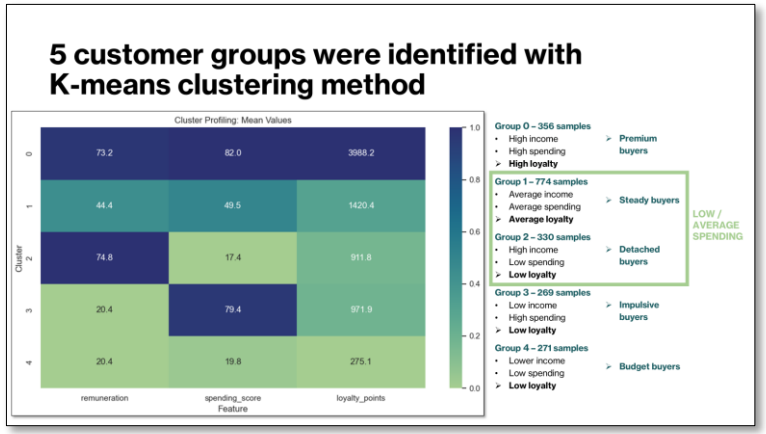
Conclusion

1. Loyalty is driven more by **behavioral factors - spending and remuneration** than demographics

2. **Target** biggest customer groups with low/average spending
   → ***Detached buyers***
   → ***Steady buyers***
   to drive loyalty

3. If financially feasible, introduce targeted, segment-specific actions also for other segments

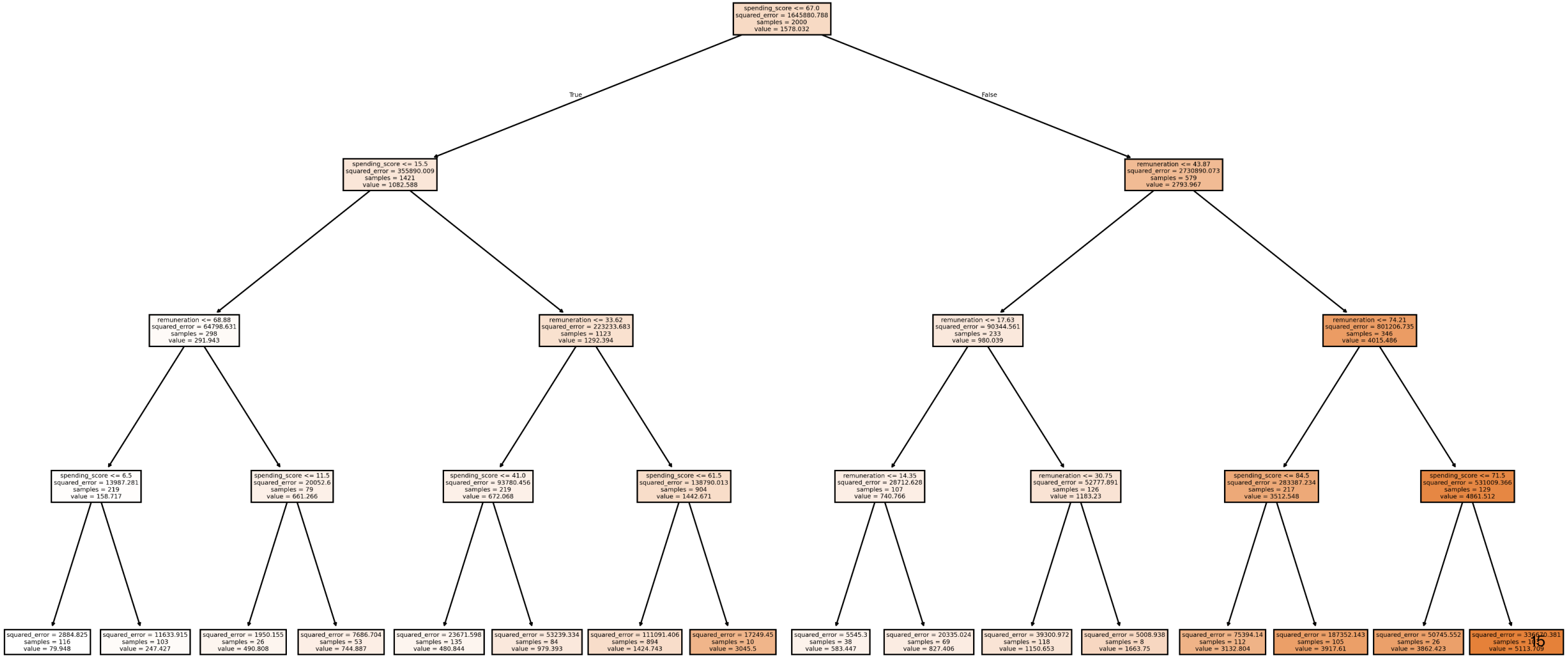
4. **Leverage sentiment insights** to guide product improvements and clearer customer messaging, supported by more refined NLP tooling



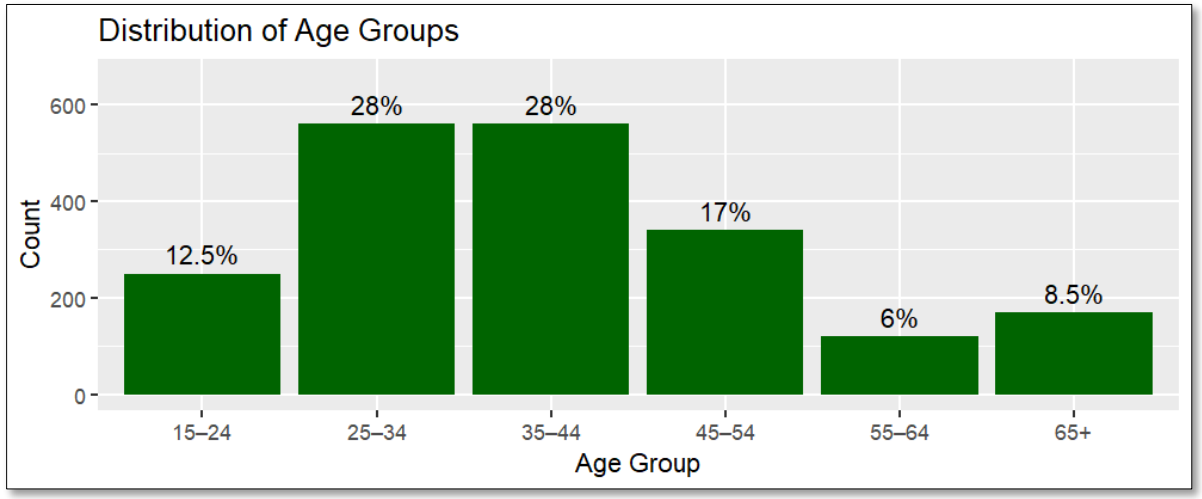
Backup

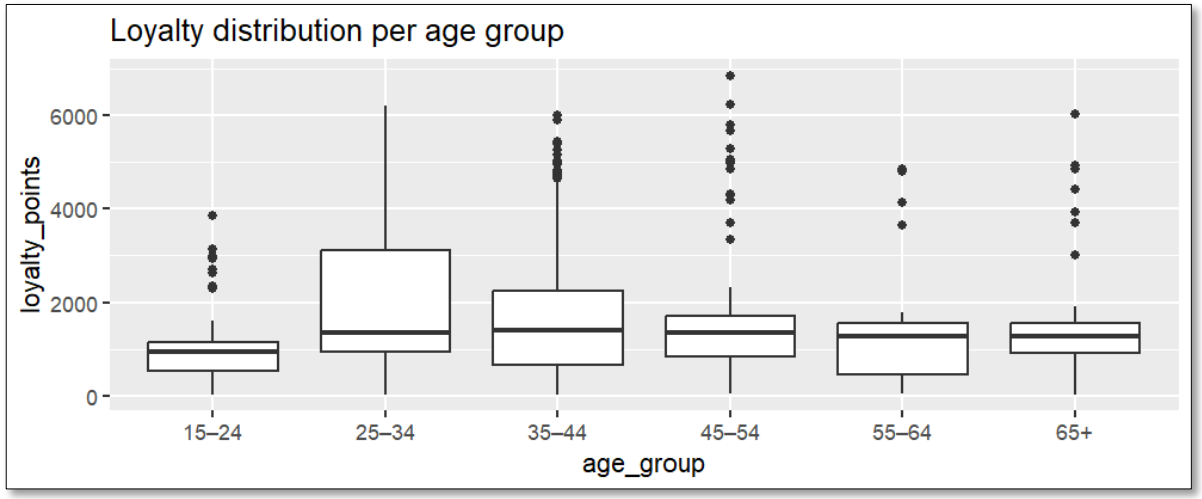
Decision tree with max depth = 4

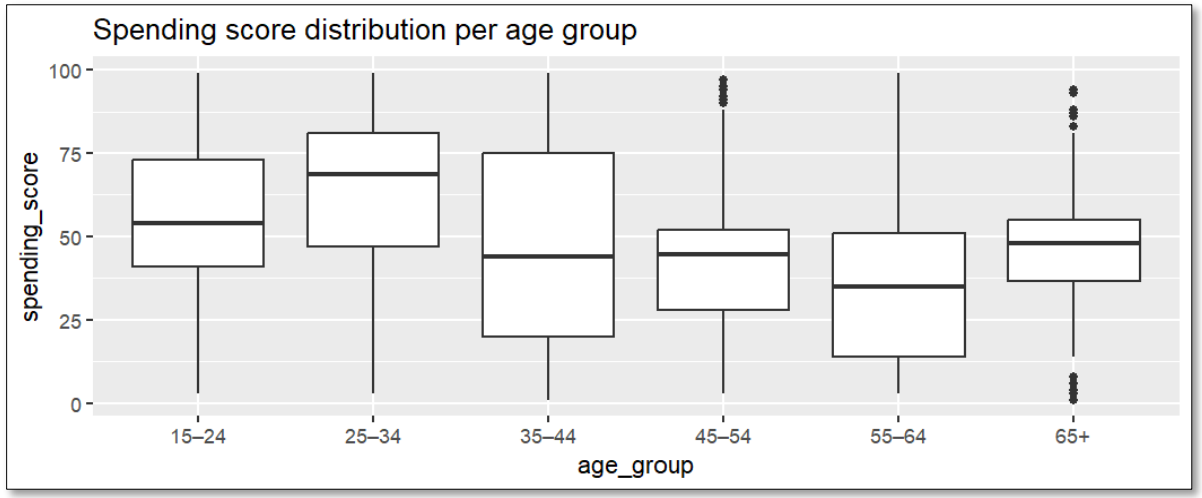
Full data set; y = loyalty points, x = remuneration, spending score

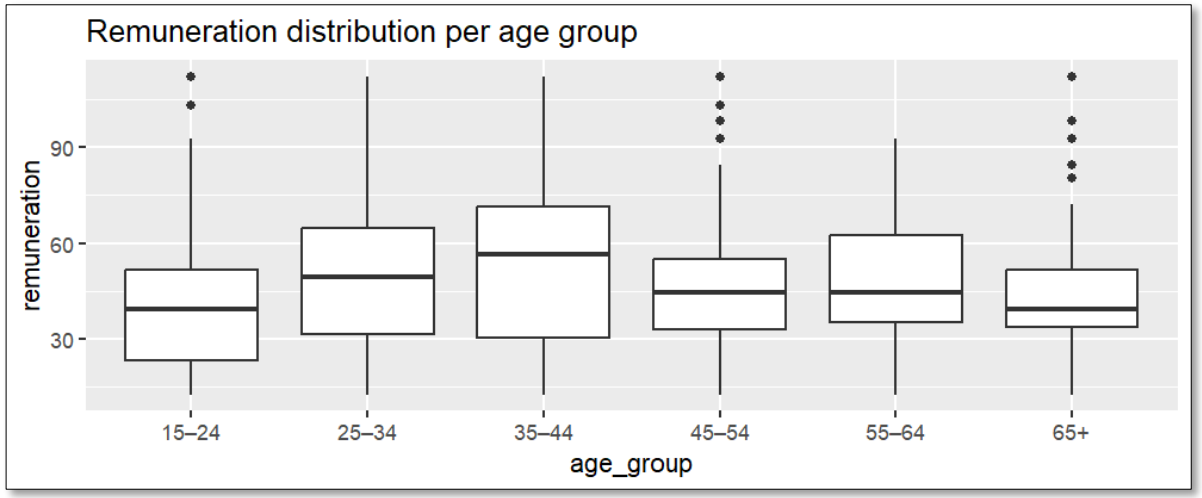


Distribution: Age vs Loyalty

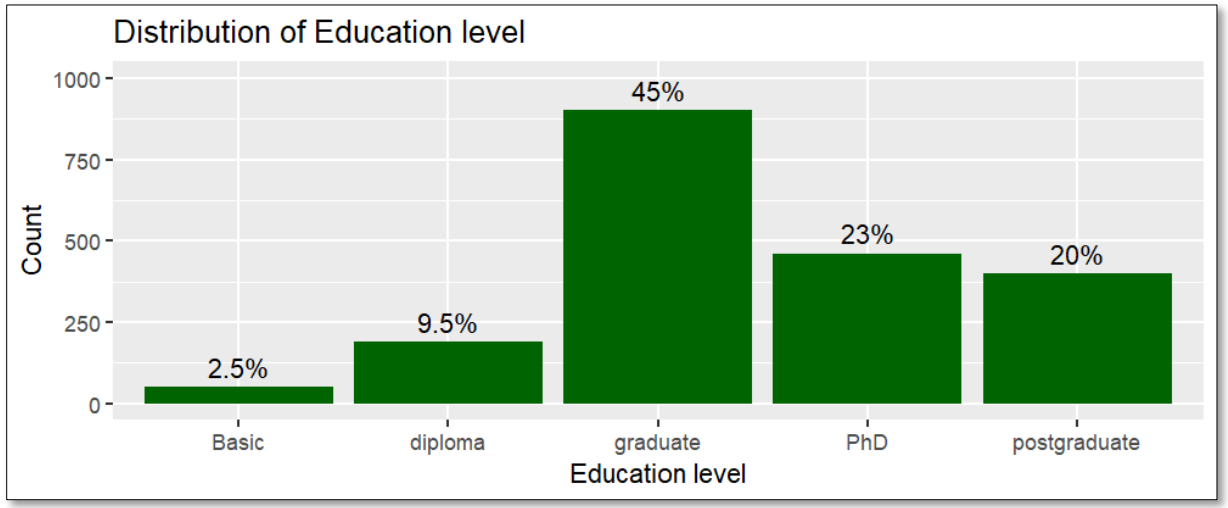


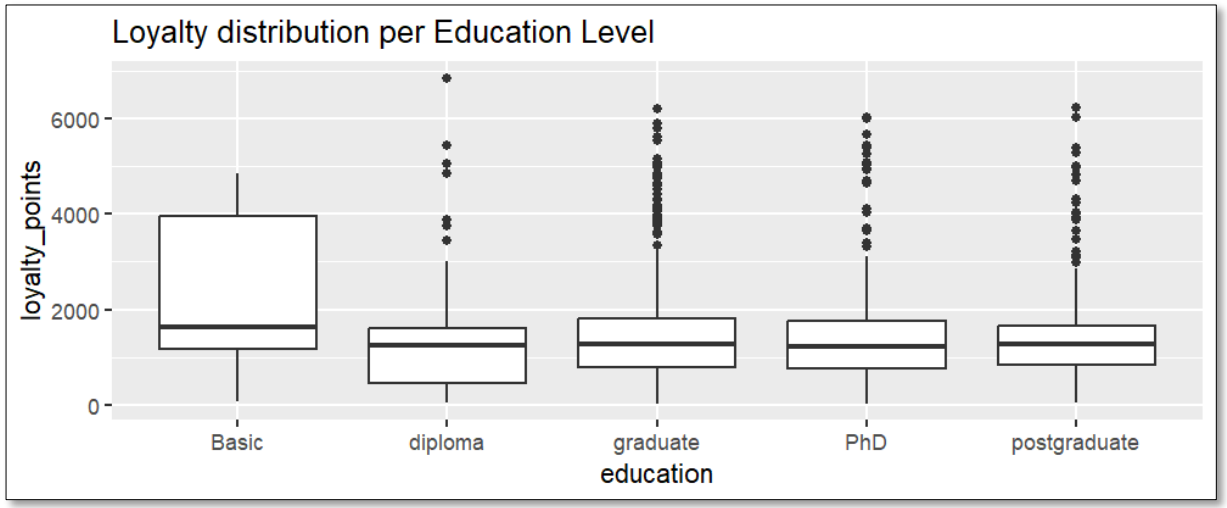


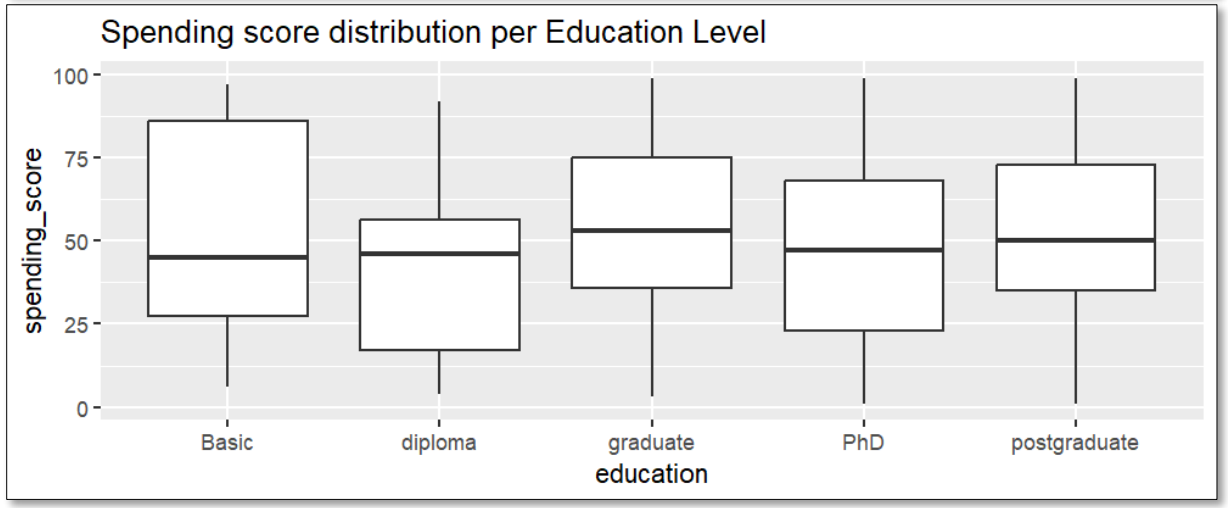


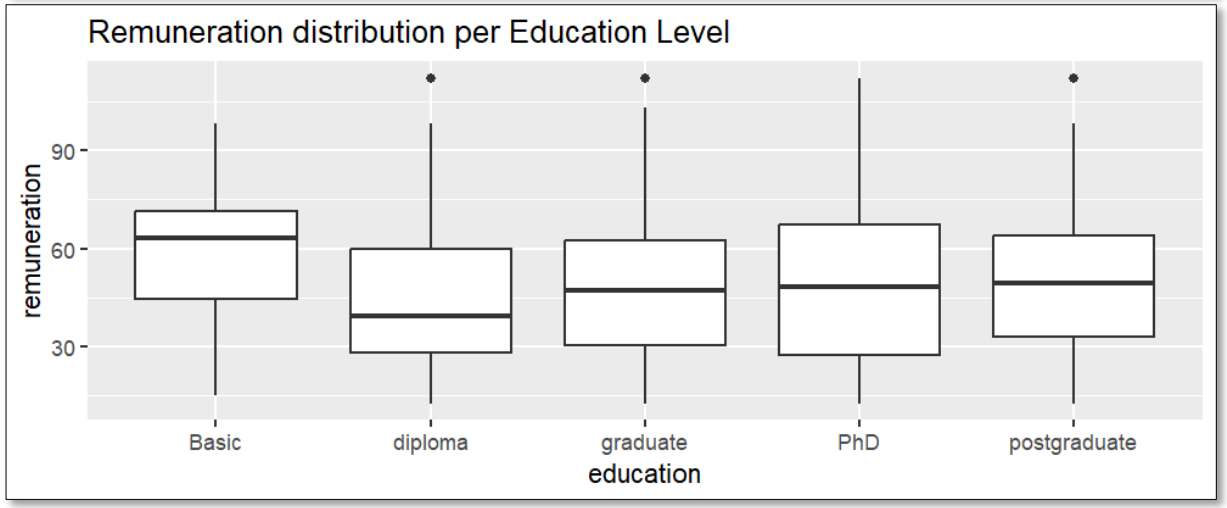


Distribution: Education vs Loyalty

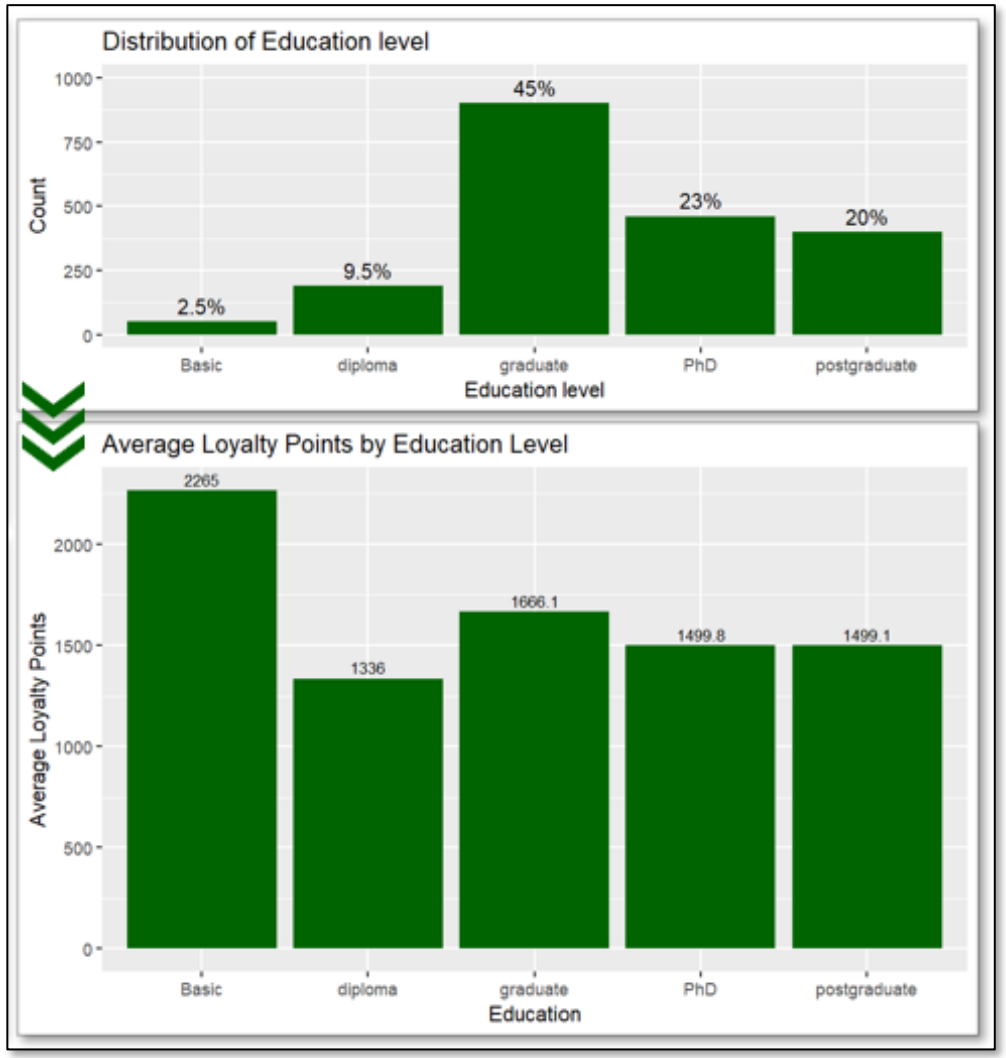








Additional insight: Education



While most customers are highly educated, **the small basic-education segment* (2.5%) shows the highest loyalty**.

This segment is underrepresented, indicating a **potential niche growth opportunity** (though not statistically significant).

→ **Priority remains on behavioral segmentation**, with spending as the primary actionable driver of loyalty

*n = 50